

Binary rank is not multiplicative under Kronecker product

Status: complete finite proof and exact positive certificate; independent author/program recheck and historical-novelty audit pending before public priority is asserted.

This note is deliberately self-contained. The search procedure that found the certificate is not part of the proof.

Result

Let A be the following 5×5 binary matrix, written explicitly as five rows:

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

For a binary matrix, rank_{bin} is the least number of all-one rectangles whose disjoint union is its set of 1 entries; equivalently it is the least r for a factorization $A = UV$ with binary U, V under ordinary integer multiplication.

The claim is

$$\text{rank}_{\text{bin}}(A) = 5, \quad \text{rank}_{\text{bin}}(A \otimes A) \leq 24,$$

and therefore

$$\boxed{\text{rank}_{\text{bin}}(A \otimes A) < \text{rank}_{\text{bin}}(A)^2}.$$

Thus binary rank is not multiplicative under Kronecker product.

A short proof that $\text{rank}_{\text{bin}}(A) = 5$

Five row-stars partition the support of A , so $\text{rank}_{\text{bin}}(A) \leq 5$.

Put

$$\ell = (-1, 1, -1, 1, 0)^T, \quad r = (-1, -1, 0, 1, 1)^T.$$

Direct calculation gives

$$\ell^T A = 0, \quad Ar = 0.$$

The submatrix on rows 1, 2, 3, 5 and columns 1, 2, 3, 4 has determinant -1 , hence the ordinary real rank of A is 4.

Assume for contradiction that $\text{rank}_{\text{bin}}(A) \leq 4$, so

$$A = UV, \quad U \in \{0, 1\}^{5 \times 4}, \quad V \in \{0, 1\}^{4 \times 5}.$$

Because $\text{rank}_{\mathbb{R}}(A) = 4$, both U and V have real rank four. Moreover

$$\text{col}(A) \subseteq \text{col}(U),$$

and both spaces have dimension four, so $\text{col}(U) = \text{col}(A)$. Since $\ell^T A = 0$, every column u of U satisfies $\ell^T u = 0$. Symmetrically,

$$\text{row}(A) \subseteq \text{row}(V),$$

and both spaces have dimension four, so $\text{row}(V) = \text{row}(A)$; since $Ar = 0$, every row v^T of V satisfies $v^T r = 0$. Thus each corresponding rank-one term obeys

$$\ell^T u = 0, \quad v^T r = 0. \quad (1)$$

Under ordinary integer multiplication,

$$A = \sum_k u_k v_k^T$$

is a sum of entrywise nonnegative binary rank-one matrices. Consequently every individual term satisfies $u_k v_k^T \leq A$ entrywise: no term can cover a zero entry of A .

Some rank-one term uv^T covers $A_{3,1} = 1$, so $u_3 = v_1 = 1$ (one-based indices).

From $\ell^T u = 0$,

$$u_2 + u_4 = u_1 + u_3 = u_1 + 1.$$

Since $A_{4,1} = 0$ and $v_1 = 1$, the support containment $uv^T \leq A$ forces $u_4 = 0$; binary-valuedness then forces $u_1 = 0$ and $u_2 = 1$.

Likewise $v^T r = 0$ gives

$$v_4 + v_5 = v_1 + v_2 = 1 + v_2.$$

Since $A_{3,5} = 0$ and $u_3 = 1$, support containment forces $v_5 = 0$; hence $v_2 = 0$ and $v_4 = 1$.

But then $u_2 = v_4 = 1$, so the same rank-one term covers $A_{2,4}$, while $A_{2,4} = 0$. Contradiction. Thus $\text{rank}_{\text{bin}}(A) = 5$.

Equivalently, the local obstruction can be checked directly: there is no binary pair (u, v) satisfying

$$\ell^T u = 0, \quad v^T r = 0, \quad uv^T \leq A, \quad u_3 = v_1 = 1.$$

The independent checker below exhausts all binary choices for u and v and verifies that the number of such pairs is zero.

The 24-rectangle certificate for $A \otimes A$

Index rows and columns of $A \otimes A$ by $0, \dots, 24$ in lexicographic order on pairs in $\{0, \dots, 4\}^2$. The following 24 all-one rectangles partition all $196 = 14^2$ ones exactly once:

#	Row set R	Column set C
1	$\{2, 7, 9\}$	$\{10, 13, 20, 23\}$
2	$\{10, 14, 20\}$	$\{1, 4, 16, 19\}$
3	$\{5, 9\}$	$\{1, 4, 11, 14, 21, 24\}$
4	$\{7, 9, 12, 14, 22\}$	$\{0, 3\}$

#	Row set R	Column set C
5	{18, 19}	{6, 8, 16, 18}
6	{13, 23, 24}	{1, 3, 16, 18}
7	{3, 23, 24}	{6, 8, 21, 23}
8	{1}	{5, 7, 9, 20, 22, 24}
9	{8}	{1, 3, 11, 13, 21, 23}
10	{2, 17}	{5, 8}
11	{0, 1, 5, 6}	{12}
12	{10, 11}	{2, 17}
13	{0}	{6, 7, 9, 11, 14, 21, 22, 24}
14	{5, 6, 20, 21}	{2, 22}
15	{16, 19, 21, 24}	{5, 9, 15, 19}
16	{4, 22}	{5, 8, 20, 23}
17	{3, 4}	{11, 13}
18	{4, 20}	{6, 9, 21, 24}
19	{15}	{6, 7, 9, 16, 17, 19}
20	{11}	{0, 4, 15, 19}
21	{6, 21, 24}	{0, 4, 20, 24}
22	{16, 20, 21}	{7, 17}
23	{1, 4, 6}	{10, 14}
24	{12, 14, 17, 22}	{15, 18}

The machine-readable copy is `certificates/binary-kronecker-seed5-self-k24.json`.

A small independent checker reconstructs the raw Kronecker matrix and verifies that every listed rectangle is contained in the support and every one-entry occurs exactly once:

```
python3 verification/verify_binary_kronecker_counterexample.py
```

The checker also verifies the two null vectors, the unimodular 4×4 minor, and exhaustively checks the local rank-one obstruction stated above. It does **not** import or trust the search program that found the certificate.

Why this belongs here

The target was selected from the EIG factorization/parallel-composition viewpoint. A product of individually minimal witness atlases always gives a product-size witness family, but EIG specifically suggests testing whether composition permits new cross-factor witness sharing that is invisible in either factor separately. Binary rank under Kronecker product is an exact finite model of that question.

That motivation is provenance only. The proof above is ordinary finite matrix and bique-partition mathematics and does not depend on EIG being correct.

Literature and novelty firewall

A targeted literature check on 2026-08-31 found the integer multiplicativity problem still stated as open in, among other sources:

- Angikar Ghosal and Andreas Karrenbauer, *Engineering Insights into Biclique Partitions and Fractional Binary Ranks of Matrices*, SEA 2025, LIPIcs 338, Article 18, DOI 10.4230/LIPIcs.SEA.2025.18;
- Michal Parnas, *Mathematical and computational perspectives on the Boolean and binary rank and their relation to the real rank*, arXiv:2601.13900 (2026).

The SEA 2025 paper disproves multiplicativity for the **fractional** binary rank and explicitly leaves the integer binary-rank question open. The present certificate concerns the integer binary rank.

No historical-priority claim is made here until an independent specialist search has ruled out an earlier unpublished, differently phrased, or insufficiently indexed counterexample.